TPC-MVP2: A Typed Global Route Decision after TPC-103–119

Certified Gates, Stopped Subroutes, Open Inputs, and the Next Minimal Tests

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Abstract

TPC-MVP1 resolved a proposed fixed-shift route into nine explicit working hypotheses H1–H9. TPC-103–119 then tested its positive resonance, generic signed, determinant–zero, physical reconstruction, endpoint, and reconnection interfaces. We now place those results in one typed directed acyclic graph. Each node records its literal object, fixed-shift quantifiers, normalization, evidence level, and status. This prevents a finite identity, a growing-family estimate, and a fixed-h_0 arithmetic saving from being interchanged.

The audit contains genuine progress. It proves, among other interfaces, an opened-atom cap, exact mass disintegration, a provenance-preserving affine quotient, signed-filter identities, post-bin determinant geometry, zero-mode duality, an exponent cone, and exact tests for physical frames, localization, cover, endpoint optimization, and native-atom reconnection. It also stops several specific subroutes: coefficient-blind distinctness, free-phase prime-square arguments, and phase-blind complete-tail closure cannot by themselves deliver the required conclusion.

The complete route nevertheless has neither a proved H1–H9 conjunction nor a theorem contradicting every allowed realization. Actual growing inputs are missing at H2–H7, the executable physical archives selected for the H6/H8 audits are absent, no alternative complete H8 intertwining is present in the audited snapshot, and the endpoint ledger is incomplete. Therefore the exact global verdict is

$$\boxed{\text{NOT-TESTABLE}}.$$

It is not GO, not a global INFEASIBLE verdict, and not evidence for a prime-pair lower bound. Local REROUTE labels remain attached only to the subroutes actually stopped. We finish with four minimal next tests: two determinant–zero exponent certificates, a complete native-atom archive, and one literal H3 block test. This is an L0/L1 route audit; no new L2 cancellation theorem is claimed.

Keywords: fixed shift; typed proof graph; route audit; endpoint budget; determinant zero mode; physical reconnection.

1 Scope and decision vocabulary

Fix a prescribed nonzero shift h_0, a smooth compactly supported weight W, and the original scalar packet $B_{h_0,\delta}(X)$ of TPC-15 [19]. TPC-MVP1 is a conditional architecture: its hypotheses H1–H9 were designed so that their conjunction, in one literal normalization, implies

$$B_{h_0,\delta}(X) = o(X),$$

which TPC-15 reconnects to the corresponding weighted fixed-shift prime-pair asymptotic [1]. The converse is not claimed; the architecture may be stronger than its scalar target.

The present paper asks a narrower administrative question:

> After TPC-103–119, can every required edge of that particular architecture be called with theorem-backed data?

It does not infer unproved arithmetic from the usefulness of the architecture.

Definition 1.1 (Evidence level)**.** An L0 result is an abstract finite or functional-analytic identity. An L1 result identifies that identity with the literal TPC object and its normalization. An L2 result is a growing, fixed-h_0, coefficient-specific arithmetic estimate with the power saving required by a hypothesis. Passing from L0 to L1, or from L1 to L2, requires an explicit theorem.

Definition 1.2 (Node record)**.** A node record is the six-tuple

$$\mathfrak{n} = (\text{object}, \text{quantifiers}, \text{normalization}, \text{provenance}, \text{evidence}, \text{status}).$$

It is *typed* only when none of these fields is inferred from an ambient, averaged-shift, surrogate, or differently normalized object.

Definition 1.3 (Statuses and route verdicts)**.** For an individual target, PROVED means that its statement is a theorem at the declared type; OPEN means that the literal target is specified but its estimate is unproved; NOT-TESTABLE means that a required archive, exponent, or compatibility input is absent, so the current artifacts cannot execute the test. A conditional interface records an exact implication whose hypotheses are still open.

For route decisions, GO requires every necessary node and the strict endpoint. REROUTE means that a named subroute is stopped while an alternative remains. INFEASIBLE requires a theorem excluding every allowed realization of the declared route. A missing theorem is neither INFEASIBLE nor evidence against the underlying number-theoretic statement.

The auxiliary label STOP ROUTE means that all other gates for one fully specified literal route are proved but its complete endpoint registry has certified value at least $1/400$. It stops that declared route only. It becomes REROUTE when a typed alternative is available, and it is not global INFEASIBLE unless every allowed route is excluded.

2 A typed decision calculus

Let $\mathcal{H} = \{H1, \dots, H9\}$ be the MVP1 gates. An evidence paper may supply an exact identity to a gate, isolate a missing growing input, or stop a particular sufficient subroute. None of those actions changes the statement of another gate.

Theorem 2.1 (Global decision rule)**.** *For the declared MVP1 architecture, use the following mutually exclusive rule.*

(i) Return GO *only if every gate in $\mathcal{H}$ has a proved typed record in one compatible quantifier scope and the actual physical ledger satisfies $\Lambda_{\mathrm{phys}} < 1/400$.*

(ii) Return INFEASIBLE *only if theorem-backed lower bounds or logical incompatibilities contradict every admissible realization of a necessary gate.*

(iii) Return REROUTE *for a selected subroute when that subroute is stopped but a typed alternative remains.*

(iv) Otherwise, if a necessary estimate or typed input is open or absent, return NOT-TESTABLE*.*

In particular, a stopped subroute cannot be promoted to global INFEASIBLE*, and an unknown endpoint token cannot be set to zero to produce* GO*.*

Proof. Item (i) is the invocation rule for the conditional synthesis theorem of MVP1. If one premise lacks a typed theorem, modus ponens is not available. Item (ii) requires universal coverage of the permitted realizations; a counterexample to one sufficient bound or one representation has weaker quantifiers. Item (iii) preserves that distinction. The remaining case is precisely the absence of enough information to execute either the upper or lower certificate, which is NOT-TESTABLE by definition 1.3. Finally, inserting zero for an unknown nonnegative loss changes the physical optimization problem rather than solving it. ∎

2.1 The dependency graph

The audit graph separates evidence bundles from gates and gates from the final conditional synthesis:

$$\begin{tikzcd}
\boxed{\text{TPC-103–109}} \arrow[d] & \boxed{\text{TPC-110–112}} & \boxed{\text{TPC-113–119}} \arrow[d] \arrow[d, shift left] \\
\{H2, H3\} \arrow[dr] & \{H5\} \arrow[d] & \{H1, H4, H6, H7, H8, H9\} \arrow[dl] \\
& \boxed{\text{typed H1–H9 conjunction at fixed } h_0} \arrow[d] & \\
& \boxed{B_{h_0,\delta}(X) = o(X) \text{ within the MVP1 route}}. &
\end{tikzcd}$$

This display is an audit DAG, not a claim that one evidence paper proves the gate below it. Every H-node is an ancestor of the final conditional target; its status is given in table 3.

3 Paper-by-paper audit: TPC-103–119

Tables 1 and 2 record the strongest exact result and the missing item that prevents overpromotion. "L1 interface" can include an exact finite identity on the literal carrier; it does not mean that the required growing estimate follows.

Table 1: Positive, generic, and determinant–zero audits.

TPC	Exact contribution	Status	Unresolved or stopped item
103	Opened-atom cap $W_X = X^{o(1)}$ on the normalized literal carrier.	PROVED, L1	The other H2 ledgers are not implied.
104	Exact principal-mass disintegration before width grouping.	L1 interface	Actual endpoint and inverse-length mass bounds are OPEN.
105	Provenance-preserving quotient by equal literal affine maps.	PROVED, L1	Growing class census U_X and distinct-map target remain OPEN.
106	Exact distinct-map character/Gram identity.	L1; REROUTE	Geometry-only distinctness is stopped; actual profile coherence remains OPEN.
107	Exact signed-filter-before-majorant identity.	PROVED, L1	A growing signed-energy bound is OPEN.
108	Exact literal-block TT^*, aliasing, and obstruction identities.	L1 interface	The H3 power saving is OPEN; a broad Chowla-strength replacement is not available.
109	Exact coherent prime-square free-phase interval and sharp saturation.	L1; REROUTE	Coefficient-blind/free-phase route is stopped; literal coefficient-specific phases remain OPEN.
110	Post-bin SVD and $D_X = E_X \alpha_X \overline{m}_X$.	PROVED, L1 identity	An actual lower bound for its three growing inputs is OPEN.
111	Exact zero-mode coarsening invariance and signed-prefix/BV duality.	PROVED, L1 identity	The growing zero-mode exponent η_Z is OPEN.
112	Exact compatibility cone $\lambda_D \leq 2\eta_Z$ and separate-reserve rule.	Conditional L1	Neither actual exponent is certified; H5 is not thereby proved.

The detailed statements appear in TPC-103–112 [2–11] and TPC-113–119 [12–18]. TPC-18 itself treats a complete global block cover as an additional hypothesis [20]; local block identities therefore cannot silently supply the executable certificates missing in TPC-117 or TPC-119.

4 Projection onto H1–H9

Theorem 4.1 (MVP2 global verdict)**.** *For the artifacts TPC-103–119 and the H1–H9 architecture stated in TPC-MVP1,*

$$\boxed{\text{Verdict}_{103:119} = \text{NOT-TESTABLE.}}$$

There is no established L2 *gate in this audit. The result is not a global* INFEASIBLE *certificate.*

Table 2: Physical, endpoint, and reconnection audits.

TPC	Exact contribution	Status	Unresolved or stopped item
113	Declared quotient condition number $\kappa_{\mathrm{q}} = \sigma_{\max}/\sigma_{\min}^{+}$ and a sharp coherence obstruction.	Conditional L1 interface	The actual H9 operator factorization, norm conversion, and growing exponent are OPEN.
114	Literal residual coarea, native maps, Gram dictionary, and collision-kernel obstruction.	L1 interface	Actual grouping census, protected complement, H9 norm conversion, and native lower transfer are OPEN.
115	Exact fixed-h_0 evaluation/Christoffel cost $K = b_0^* G^{\dagger} b_0$.	L1 interface	Actual profile subspace and growing localization cost are OPEN.
116	Exact conditional closure theorem and phase-blind obstruction.	L1; local REROUTE	Complete outer tail bound is OPEN; phase-blind closure is stopped.
117	Exact full-cover test $P = BB^{\dagger}$, residual $(I - P)w$.	NOT-TESTABLE for H6	The actual growing B, w archive and residual estimate are absent.
118	Nonduplicated physical ledger and exact rational primal–dual endpoint logic.	INCOMPLETE	Several actual growing costs are unknown; no strict endpoint certificate exists.
119	Native-atom conservation and a sufficient canonical reconnection-matrix criterion.	NOT-TESTABLE for its H8 audit	The dated snapshot has neither the selected complete leaf archive nor another complete exact intertwining joining every TPC-15 atom to the final synthesis.

Table 3: Current status of every required MVP1 gate.

Gate	Literal role	Status	Reason
H1	Canonical physical synthesis	NOT-TESTABLE	Exact frame and native Gram criteria exist, but no complete actual-carrier frame/archive or proved growing cost is supplied.
H2	Literal positive resonance	OPEN	The atom cap is proved; principal/cross-map/coherence ledgers are not. Some geometry-only subroutes are stopped.
H3	Restricted signed affine arithmetic	OPEN	The literal TT^* object is isolated, but no fixed-h_0 $X^{-1/200+o(1)}$ correlation saving is proved.
H4	Complete high/ultra return	OPEN	Conditional closure and an obstruction are known; the complete original-normalization outer bound is not.
H5	Determinant/zero compatibility	NOT-TESTABLE	The cone is exact, but the actual λ_D, η_Z inputs are absent.
H6	Full-block physical return	NOT-TESTABLE	The exact projector test cannot be run without the growing physical B, w archive.
H7	Prescribed-h_0 localization	OPEN	The exact Christoffel cost is known; the literal profile and subcritical growing bound are not.
H8	Original-packet reconnection	NOT-TESTABLE	Native conservation is exact, but the sufficient proof-carrying leaf matrix selected in TPC-119 is absent and the audited snapshot contains no alternative complete exact intertwining.
H9	Strict endpoint	NOT-TESTABLE	The LP logic is exact; the actual token registry is incomplete, so $\Lambda_{\mathrm{phys}} < 1/400$ is uncertified.

Proof. Table 3 contains four explicitly open gates and five gates whose actual test lacks a required typed input. Hence the premise of theorem 2.1(i) fails. TPC-106, TPC-109, and TPC-116 stop only coefficient-blind, free-phase, or phase-blind sufficient subroutes; coefficient-specific signed routes remain logically available. Thus the universal premise of theorem 2.1(ii) is absent. Applying item (iv) gives NOT-TESTABLE. None of the papers listed in tables 1 and 2 supplies the fixed-shift growing cancellation demanded in definition 1.1, so no L2 gate is established. □

Remark 4.2 (What has moved forward). The verdict does not mean "no progress." Before these audits, an apparent failure could be hidden among normalization, aliasing, coherence, zero mode, cover, or reconnection. The current graph gives exact tests for those interfaces and identifies which data must be measured or proved. That is structural L1 progress. It is not fixed-h_0 parity cancellation.

5 The strict endpoint is an independent stop gate

TPC-118 represents every physical reconstruction event by one amplitude-loss token, merges exact duplicates, and replaces constituent tokens only when a joint theorem actually subsumes them [17]. On a fixed affine route cell the decision is an exact rational linear program with optimum Λ_*. Its interpretation is

$$\begin{array}{rcl} \Lambda_* < 1/400 & \Longrightarrow & \text{GO for a complete theorem-backed ledger,} \\ \Lambda_* = 1/400 & \Longrightarrow & \text{equality stop,} \\ \Lambda_* > 1/400 & \Longrightarrow & \text{stop that route,} \\ \text{required token unknown} & \Longrightarrow & \text{NOT-TESTABLE.} \end{array}$$

Proposition 5.1 (No zero-filling and no reserve reuse)**.** *The current endpoint cannot be certified by either*

(a) assigning exponent 0 *to an unknown frame, grouping, localization, tail, or cover cost; or*

(b) spending slack from the determinant–zero condition $\lambda_D \le 2\eta_Z$ *as physical endpoint slack.*

Moreover, equality $\Lambda_{\text{phys}} = 1/400$ *does not pass H9.*

Proof. Operation (a) changes an incomplete physical ledger into a different optimization problem. Operation (b) merges two ledgers that TPC-112 and MVP1 declare independent absent a new intertwining theorem [1, 11]. Finally, the synthesis starts from a packet saving of $1/400$; equality consumes the whole fixed-power reserve and leaves only an $X^{1+o(1)}$ upper bound, which does not certify $o(X)$. □

6 The next four minimal tests

The decision graph recommends small falsifiable gates rather than a new global claim.

TPC-121: actual post-bin fiber phase/participation. TPC-110 gives

$$D_X = E_X \alpha_X \overline{m}_X.$$

The next determinant test should preserve the actual post-bin fibers and certify the diagonal energy scale, the coefficient-specific phase/kernel angle, and effective fiber participation. Success would supply a candidate λ_D; failure could stop that coherence mechanism without stopping every H5 route.

TPC-122: signed-prefix/BV/content transfer. Use the exact Abel/BV duality of TPC-111 on the literal ordered fibers. A theorem must retain the arithmetic sign and fixed h_0, bound signed prefixes, charge the reconstruction BV envelope, and control the content remainder. Its output is a candidate η_Z, to be compared with $\lambda_D/2$, not a generic nonzero-frequency spectral gap.

TPC-123: complete native-atom archive. Independently of H5, instantiate the TPC-119 canonical matrix: every opened atom (ℓ, k, d) must have one recoverable provenance key, every retained or soft leaf must appear, and column conservation must hold at the same h_0 and normalization. A failed audit locates the first missing, duplicated, rescaled, or surrogate branch.

TPC-124: one literal H3 block. Select one actual long generic affine block and preserve its Möbius weights, r-dependent phase, affine character, reconstruction multiplier, all prefixes, and prescribed shift. Compute the exact normalization and test a rigorous upper/lower interval for its signed TT^* correlation. A finite experiment can choose the next analytic lemma, but only a growing uniform theorem would count as L2.

Corollary 6.1 (Interpretation of the next queue)**.** *TPC-121 and TPC-122 attack the two missing exponent inputs of H5. TPC-123 attacks the H1/H6/H8 physical archive, while TPC-124 attacks H3 directly. Progress on one branch may be published independently, but none substitutes for the others in the H1–H9 conjunction.*

7 Reproducible audit and claim boundary

The standard-library script `experiments/tpc120_global_route_audit.py` encodes the paper-to-gate edges and verifies:

1. all nine H-nodes occur and have evidence ancestors;
2. no TPC-103–119 record is labeled as a proved L2 result;
3. open or absent required inputs return NOT-TESTABLE;
4. a local stopped subroute is not global INFEASIBLE;
5. strict-below, equality, strict-above, and incomplete endpoint examples have four different outcomes;
6. unknown costs are not converted into a zero-cost certificate; and
7. the next queue is exactly TPC-121–124.

The generated JSON is a regression artifact for the decision logic. It is not a computation of any prime or Möbius correlation.

Established. The status calculus, typed dependency graph, exact endpoint decision rule, and paper-to-gate audit are L0/L1. The cited papers contain the individual exact identities summarized above.

Not established. This paper proves no H1–H9 conjunction, no strict actual-carrier $\Lambda_{\text{phys}} < 1/400$ bound, no $B_{h_0,\delta}(X) = o(X)$ estimate, and no new L2 fixed-shift cancellation. It does not break the parity barrier, prove a Hardy–Littlewood asymptotic, produce a prime-pair lower bound, or prove the twin-prime conjecture. Its conclusion is a route decision: the path still has typed alternatives, but the current archive is insufficient to decide their success.

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